

SHARP: Spectrum-aware Highly-dynamic Adaptation for Resolution Promotion in Remote Sensing Synthesis

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Abstract

Text-to-image synthesis for remote sensing (RS) holds immense potential to advance Earth observation tasks, yet the community lacks an accessible, high-performance generative framework. A critical bottleneck is the inherent demand for large-scale, high-resolution imagery, where directly training foundational models incurs prohibitive computational and temporal costs. While achieving training-free resolution promotion via Rotary Position Embedding (RoPE) extrapolation offers a cost-effective alternative, current algorithms rely on static scaling rules that are heavily biased toward natural scenes. Natural images typically feature single, salient subjects, whereas RS imagery is characterized by dense, scattered, and minute instances that are highly dependent on high-frequency structural integrity. To overcome this fundamental domain mismatch, we present a comprehensive framework for large-scale RS image synthesis. First, we establish a robust domain-specific generative prior by training on a curated dataset of over 100,000 RS images. Second, to accomplish high-fidelity resolution promotion, we propose a novel training-free positional extrapolation algorithm, named **SHARP**. At its core, SHARP introduces a rational decay scheduler, $\kappa_{rs}(t)$, to continuously modulate RoPE frequencies step-by-step throughout the entire denoising process. By explicitly coupling the extrapolation strength to the evolving spectral state of diffusion—enforcing strong global layout constraints early on and progressively relaxing them to recover dense high-frequency details later—SHARP preserves crisp instance features while seamlessly expanding the spatial footprint. Extensive experiments demonstrate that our approach consistently achieves state-of-the-art performance across multiple promoted resolutions with negligible overhead, providing a highly efficient and structurally faithful solution for large-scale RS generation. Our model weights and code are fully open-sourced at <https://github.com/bxuanz/SHARP>.

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Image manipulation*.

Keywords

remote sensing image synthesis, diffusion transformers, resolution promotion, rotary position embedding, high-resolution generation

1 Introduction

Remote sensing (RS) imagery [1, 8, 26, 33, 34, 37] is central to Earth observation tasks, yet the community lacks an accessible, high-performance generative foundation for text-to-image synthesis [14, 30, 31, 35]. A critical bottleneck in advancing this field is the

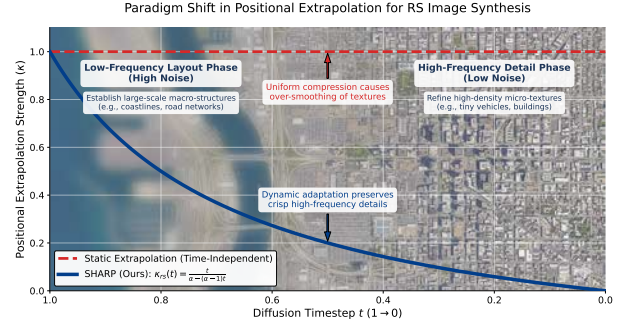


Figure 1: Static vs. dynamic positional extrapolation for resolution promotion. Diffusion-based generation follows a frequency-progressive paradigm: global layouts emerge at high noise levels while dense high-frequency details crystallize at low noise levels. Static methods (red dashed) apply uniform extrapolation strength across all steps, uniformly compressing the spatial spectrum and causing catastrophic over-smoothing of fine remote sensing structures. SHARP (blue solid) achieves highly-dynamic adaptation by coupling the extrapolation strength with the denoising progression via a rational decay scheduler $\kappa_{rs}(t)$, continuously preserving crisp high-frequency details without sacrificing structural coherence.

inherent demand for large-scale, high-resolution imagery. A single generated RS image must span a vast spatial extent while preserving a massive amount of dense, scattered, and tiny local structures (e.g., vehicles, complex road networks) [15, 17]. Directly training foundational Diffusion Transformers (DiTs) [18] at such extreme resolutions incurs prohibitive computational costs due to the quadratic scaling of self-attention. Therefore, achieving training-free resolution promotion at inference time—generating at sizes vastly beyond the training resolution (e.g., $1024 \times 1024 \rightarrow 2048 \times 2048$)—is highly desirable.

Current training-free resolution promotion approaches predominantly rely on *positional extrapolation*, specifically by rescaling Rotary Position Embedding (RoPE) [28] frequencies (e.g., PI [5], NTK-aware [4], YaRN [19]). However, these algorithms were designed for Large Language Models (LLMs) or natural images, which typically feature single salient subjects and smooth textures. In stark contrast, RS imagery is overwhelmingly dominated by high-frequency structural details. Directly migrating these methods to RS generation exposes a severe domain mismatch.

Crucially, existing training-free algorithms rely on a *static* extrapolation rule that remains fixed throughout the entire denoising process. This rigid assumption fundamentally conflicts with the inherent generative physics of diffusion models [7, 11, 20, 21, 27, 36]. Diffusion-based image synthesis follows a coarse-to-fine, frequency-progressive paradigm: during the initial high-noise phase, the model prioritizes converging on the low-frequency *global layout* (e.g., broad coastlines, primary road networks); subsequently, in the late low-noise phase, the generative focus shifts to crystallizing high-frequency *texture details* (e.g., dense vehicles, minute building edges).

Current algorithms apply a uniform, unyielding positional compression from beginning to end (red dashed line in Fig. 1). While this static extrapolation strength is necessary early on to establish an expanded layout, it acts as a severe spectral bottleneck during the later detail-refinement phase. By continuously compressing the spatial frequencies precisely when delicate micro-textures are trying to emerge, static methods irreparably suppress high-frequency components. For RS imagery, this leads to catastrophic over-smoothing, washing out the critical structural integrity of fine instances.

To overcome this fundamental domain mismatch and computational bottleneck, we present a comprehensive framework for large-scale RS text-to-image synthesis. First, we establish a robust domain-specialized prior (**RS-FLUX**) by fine-tuning FLUX [3] on a curated dataset of over 100,000 high-quality RS images. Second, to accomplish high-fidelity resolution promotion, we propose **SHARP** (Spectrum-aware Highly-dynamic Adaptation for Resolution Promotion).

Unlike static baselines, SHARP achieves *highly-dynamic adaptation* by introducing a rational decay scheduler $\kappa_{rs}(t)$ (blue solid line in Fig. 1) that continuously modulates RoPE frequencies step-by-step. Our core insight is to explicitly synchronize the positional extrapolation strength with the natural spectral evolution of the diffusion process. During the early global-layout generation phase, SHARP applies strong positional extrapolation to guarantee macro-layout coherence across the expanded spatial canvas. As the generation transitions into the texture-refinement phase, SHARP smoothly and continuously decays the extrapolation strength. By deliberately reducing the positional compression when precise textures are forming, SHARP liberates the model to render crisp micro-structures without spatial distortion. This dynamic decoupling seamlessly expands the image footprint while faithfully preserving intricate RS details.

In summary, our main contributions are:

- **Domain-Specialized Generative Prior:** We construct a 102,952-pair RS text-to-image dataset and fine-tune FLUX to build **RS-FLUX**, providing a powerful foundational DiT for the RS community.
- **Identifying the Static Bottleneck:** We identify and theoretically analyze the fatal conflict between existing *static* positional extrapolation and the frequency-progressive nature of diffusion, revealing the root cause of over-smoothing in high-frequency RS generation.
- **Dynamic Extrapolation Algorithm (SHARP):** We propose **SHARP**, a novel training-free algorithm that dynamically couples RoPE frequencies with the denoising spectrum.

Extensive experiments demonstrate that SHARP effectively eliminates over-smoothing, achieving state-of-the-art performance across multiple promoted resolutions with negligible overhead.

2 Related Work

2.1 Remote Sensing Image Generation

Generative modeling for RS imagery has evolved significantly, transitioning from GAN-based methods to highly expressive diffusion models [9, 11]. While early RS generative research demonstrated the value of diffusion priors, they predominantly targeted specific restoration or conditional tasks (e.g., super-resolution [22], cloud removal [29]). For open-ended text-to-image synthesis, most existing RS adaptations continue to rely on U-Net backbones [24], which are inherently biased toward patch-level synthesis rather than holistic, complex scene generation. The recent success of Diffusion Transformers (DiTs) [3, 18] in producing photorealistic natural images by scaling self-attention motivates our development of **RS-FLUX**, a DiT-based generative foundation model tailored for the dense and complex structural statistics of RS imagery.

2.2 Positional Extrapolation in Diffusion Transformers

Training foundational DiTs at extreme resolutions is computationally prohibitive. Consequently, training-free resolution promotion—generating images at scales vastly exceeding the training context window—has become essential. In modern DiTs, Rotary Position Embedding (RoPE) [28] is the standard spatial encoding mechanism. For a token at position m , RoPE encodes positional information by rotating the hidden representations using a set of fixed frequencies:

$$\theta_i = b^{-2i/d} \quad (1)$$

where d is the feature dimension, $i \in [0, d/2)$ is the channel index, and b is the base wavelength (typically 10000). The corresponding rotation angle applied to the feature space is $m\theta_i$.

When promoting resolution at inference time, the positional index m exceeds the context window seen during training, leading to severe out-of-distribution (OOD) degradation. Representative training-free methods mitigate this by rescaling the position or frequencies. Position Interpolation (PI) [5] linearly compresses positions to m/s (where s is the resolution promotion factor), effectively compressing all frequencies uniformly. NTK-aware scaling [4] modifies the base wavelength b to preserve high-frequency information, while YaRN [19] applies a more sophisticated, frequency-selective ramp. While computationally lightweight, these extrapolation techniques were originally designed for Large Language Models (LLMs) or natural images, overlooking the unique structural demands of the RS domain.

2.3 The Bottleneck of Static Extrapolation

A fundamental limitation shared by PI, NTK-aware scaling, and YaRN is their reliance on a *static* extrapolation strategy: the exact same frequency scaling modification is applied to θ_i uniformly across every denoising timestep t .

This static assumption severely conflicts with the frequency-progressive generative physics of the diffusion process [11, 27],

where high-noise early steps construct the global layout (low-frequency) and low-noise terminal steps crystallize dense details (high-frequency). For RS imagery, which is defined by an abundance of high-frequency spatial energy (e.g., vehicles, complex building contours), this temporal mismatch is fatal. Uniformly compressing the positional spectrum $m\theta_i$ at the late stages of denoising directly suppresses the high-frequency components critical for resolving fine instances, inevitably leading to over-smoothed outputs. SHARP fundamentally resolves this by breaking the static assumption, introducing a dynamically modulated promotion schedule that respects the instantaneous spectral state of the diffusion process.

3 Methodology

This section presents our two-component framework. Section 3.1 describes RS-FLUX, the domain-specialized DiT prior obtained by fine-tuning FLUX [3] on curated RS data. Section 3.2 establishes the spectral motivation that directly informs the design of SHARP (Sec. 3.3), our training-free dynamic resolution promotion strategy. The overall pipeline is illustrated in Fig. 4.

3.1 Domain-Specialized Prior: RS-FLUX

Open-source DiT checkpoints [18] are optimized for natural images and fail to capture the semantics and spectral statistics of aerial scenes. RS imagery contains dense man-made structures, small objects, and repetitive textures under-represented in natural-image priors [6]. Parameter-efficient approaches such as LoRA [12] constrain updates to a low-rank subspace that may be insufficient for bridging the large distribution gap between natural and RS imagery. We therefore perform full fine-tuning of FLUX [3] on over 100,000 high-quality RS images, updating all transformer parameters so that attention heads, feed-forward layers, and the final projection jointly adapt to the spectral and geometric statistics unique to overhead imagery. The resulting RS-FLUX serves as the base generator in all experiments. Because SHARP operates exclusively at inference time by modifying the RoPE frequency table, it requires no additional high-resolution data or retraining—the two components are fully decoupled.

3.2 Spectral Motivation and Frequency-Progressive Denoising

To motivate the dynamic promotion strategy, we first quantify the spectral gap between natural and RS images. Following the protocol in Fig. 2, we compute the average normalized radial PSD over 100 ImageNet [6] and 100 DIOR [15] images. After low-frequency normalization, RS imagery retains systematically stronger medium- and high-frequency energy (Fig. 2). Uniform spectral compression is therefore more damaging for RS than for natural-image generation.

This empirical analysis establishes *what* differs between the two domains; we next analyze *when* different frequency bands become recoverable. Following the flow-matching formulation [16] used by FLUX [3], the noisy latent at normalized time $t \in [0, 1]$ is

$$x_t = (1 - t)x_0 + t\epsilon, \quad (2)$$

where $x_0 \sim p_{\text{data}}$ and $\epsilon \sim \mathcal{N}(0, I)$. Applying the Fourier transform yields

$$\hat{x}_t = (1 - t)\hat{x}_0 + t\hat{\epsilon}. \quad (3)$$

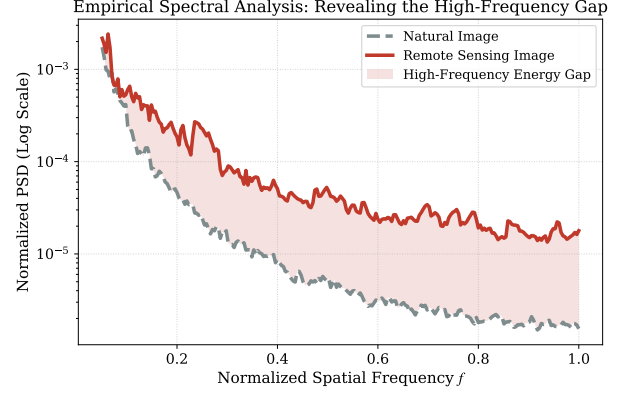


Figure 2: Empirical Spatial Spectrum Analysis. Average normalized radial power spectral density (PSD) over 100 ImageNet [6] and 100 DIOR [15] images. Each image is converted to grayscale, resized to 512×512 , windowed with a 2-D Hanning function, transformed via centered FFT, radially averaged, and normalized by its non-DC low-frequency baseline. RS imagery exhibits systematically stronger medium- and high-frequency energy.

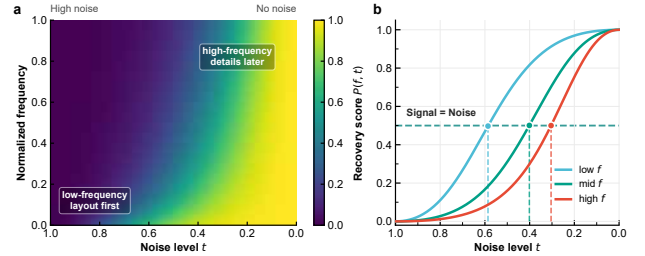


Figure 3: Frequency-Progressive Denoising. Left: heatmap of the bounded recovery score $P(f, t) = \rho(f, t)/(1 + \rho(f, t))$ over normalized frequency and reverse denoising progress. Lower-frequency bands enter the signal-dominant regime earlier. Right: representative trajectories for low-, medium-, and high-frequency components, whose threshold crossings occur progressively later, motivating strong early promotion and weaker late extrapolation.

Assuming independence between $\hat{x}_0$ and $\hat{\epsilon}$ with white noise spectrum, the expected PSD at spatial frequency f is

$$\mathbb{E}[\|\hat{x}_t\|_f^2] = (1 - t)^2 S(f) + t^2 C_\epsilon, \quad (4)$$

where $S(f) = \mathbb{E}[\|\hat{x}_0\|_f^2]$ is the clean data PSD and C_ϵ is a frequency-independent noise constant.

Each frequency band is governed by a competition between the clean signal and the noise floor. We define the signal-dominance ratio

$$\rho(f, t) = \frac{(1 - t)^2 S(f)}{t^2 C_\epsilon}. \quad (5)$$

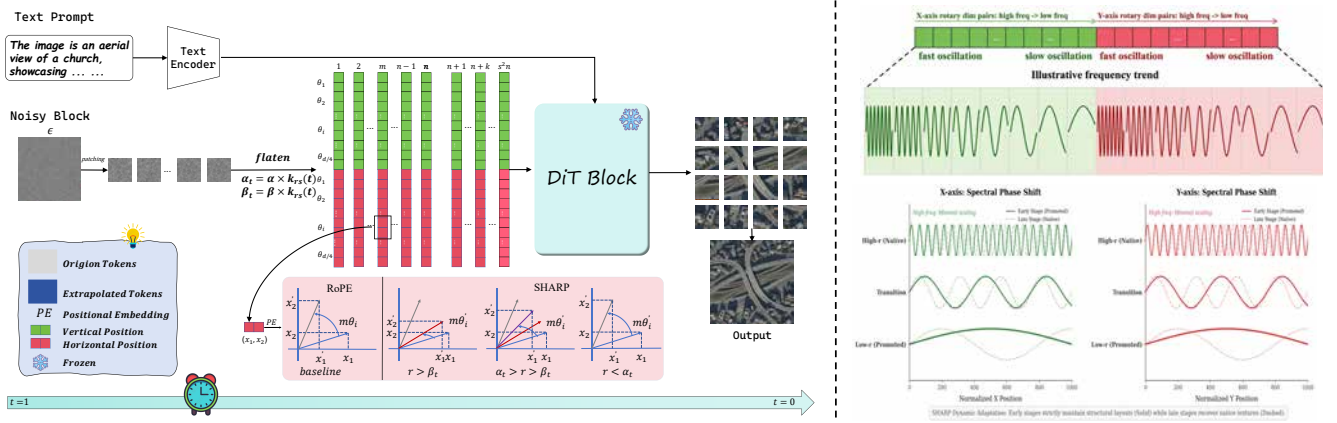


Figure 4: SHARP Overview. SHARP performs dynamic resolution promotion in RoPE through the rational decay scheduler $\kappa_{rs}(t)$ and the ramp function $\gamma(r, t)$. Strong promotion is preserved in the early layout stage and progressively reduced in the late detail-recovery stage, avoiding the over-smoothing caused by static extrapolation.

Since $S(f)$ decreases with frequency, for any $f_l < f_h$ we have $\rho(f_l, t) > \rho(f_h, t)$, meaning lower-frequency bands become signal-dominant earlier. For visualization, we define the bounded recovery score

$$P(f, t) = \frac{\rho(f, t)}{1 + \rho(f, t)} \in [0, 1], \quad (6)$$

where larger values indicate greater signal dominance. Fig. 3 visualizes this quantity over normalized frequency and reverse denoising progress.

The critical transition time $t_c(f)$, at which signal and noise are equally strong ($\rho(f, t_c) = 1$), is defined as:

$$t_c(f) = \frac{\sqrt{S(f)}}{\sqrt{S(f)} + \sqrt{C_\epsilon}}. \quad (7)$$

Because $t_c(f)$ monotonically increases with $S(f)$, and $S(f)$ naturally decays at higher frequencies, the critical time for high-frequency components is strictly smaller (i.e., closer to $t = 0$) than for low-frequency ones. In the reverse flow-matching process (from $t = 1$ to $t = 0$), this mathematical property dictates that low-frequency global layouts emerge early, while high-frequency micro-details are exclusively recoverable in the terminal stage. Combined with the empirical finding that RS imagery is heavily bottlenecked by high-frequency energy (Fig. 2), this frequency-progressive behavior provides a compelling two-fold motivation for dynamic resolution promotion: RS images are both more reliant on high-frequency fidelity and more vulnerable to its suppression by static extrapolation. Driven by this crucial insight, we propose SHARP, a dynamic resolution promotion strategy designed to explicitly couple the extrapolation strength with this instantaneous spectral state.

3.3 SHARP: Dynamic Resolution Promotion

The spectral formulation in Sec. 3.2 reveals a fundamental limitation of existing static extrapolation methods [4, 5, 19]. A time-invariant scaling rule is beneficial only during the early high-noise stage, where expanding the receptive field helps organize the global layout. However, it becomes highly detrimental in the late low-noise stage, where the exact same positional compression irreversibly

Algorithm 1 SHARP Inference

Require: Trained RS-FLUX model $\mathcal{F}$; text prompt c ; target resolution L_{target} ; training resolution L_{train} ; scheduler coefficient α_s ; transition bounds α, β ; denoising timesteps $\{t_n\}_{n=0}^N$ with $t_0 = 1, t_N = 0$

Ensure: Generated image $\hat{x}_0$ at resolution L_{target}

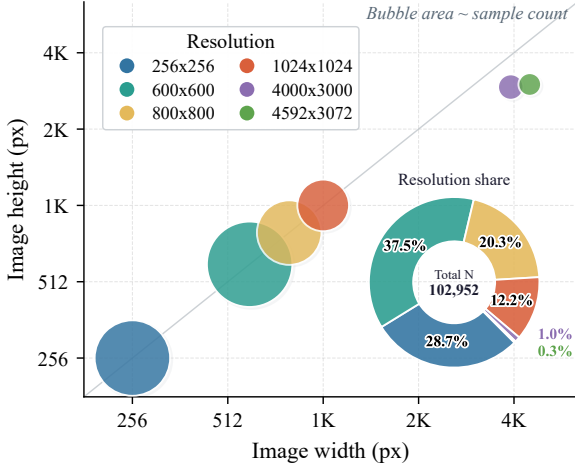
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1: $s \leftarrow L_{\text{target}}/L_{\text{train}}$ $\triangleright$ Resolution promotion factor
2: $x_{t_0} \sim \mathcal{N}(0, I)$ $\triangleright$ Initialize noise at target resolution
3: for $n = 0$ to $N - 1$ do
4: $\kappa_{rs}(t_n) \leftarrow t_n / [\alpha_s - (\alpha_s - 1) t_n]$ $\triangleright$ Rational Decay Scheduler
5: $\alpha_{t_n} \leftarrow \alpha \cdot \kappa_{rs}(t_n); \quad \beta_{t_n} \leftarrow \beta \cdot \kappa_{rs}(t_n)$ $\triangleright$ Dynamic bounds
6: for each RoPE dimension i do
7: $r(i) \leftarrow L_{\text{target}}/\lambda_i$
8: $\gamma \leftarrow \text{clamp}((r(i) - \alpha_{t_n})/(\beta_{t_n} - \alpha_{t_n}), 0, 1)$ $\triangleright$ Dynamic
ramp
9: $h_{t_n}(\theta_i) \leftarrow (1 - \gamma) \theta_i/s + \gamma \theta_i$ $\triangleright$ Rescaled frequency
10: end for
11: Apply $\{h_{t_n}(\theta_i)\}_i$ to RoPE in $\mathcal{F}$
12: $x_{t_{n+1}} \leftarrow \text{Denoise}(\mathcal{F}, x_{t_n}, t_n, c)$ $\triangleright$ One denoising step
13: end for
14: $\hat{x}_0 \leftarrow \text{Decode}(x_{t_N})$ $\triangleright$ VAE decode
15: return $\hat{x}_0$

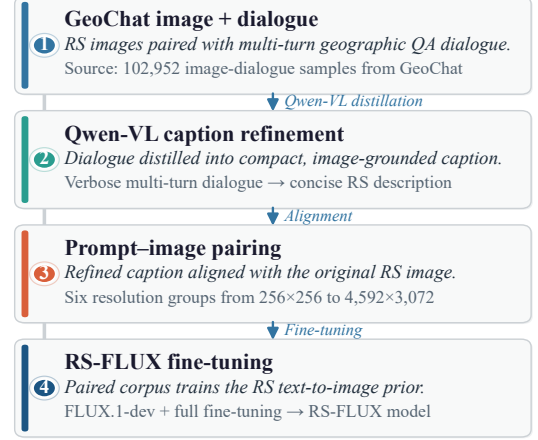
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suppresses the high-frequency components critical for RS detail recovery.

SHARP fundamentally resolves this bottleneck through a spectrum-aware, dynamically modulated promotion mechanism operating directly within the RoPE space. As illustrated in Fig. 4, at each denoising step  $t$ , SHARP dynamically calibrates the RoPE frequency table via two tightly coupled modules: (i) the *Rational Decay Scheduler* (RDS), which governs the temporal evolution of the promotion strength in alignment with the frequency-progressive recovery order (Eq. 7); and (ii) the *Dynamic Ramp Function*, which partitions the RoPE dimensions based on their effective spatial frequencies and applies differential extrapolation scaling across the frequency axis.



(a) Resolution distribution.



(b) Corpus construction pipeline.

**Figure 5: Overview of the 102,952-sample RS training corpus. (a) Resolution distribution: bubble position and area denote dimensions and sample counts, respectively, with a composition summary in the inset. (b) Corpus construction: GeoChat image-dialogue pairs are refined via Qwen-VL into descriptive captions for RS-FLUX fine-tuning.**

**Rational Decay Scheduler (RDS).** The RDS maps the normalizing denoising timestep  $t \in [0, 1]$  to a dynamic decay coefficient:

$$\kappa_{rs}(t) = \frac{t}{\alpha_s - (\alpha_s - 1)t}, \quad \alpha_s \geq 1, \quad (8)$$

where  $\alpha_s$  is the scheduler coefficient controlling the decay rate. This rational function is designed to smoothly decay from near-unity to zero. During early denoising steps ( $t \approx 1$ ),  $\kappa_{rs}(t) \approx 1$ , maintaining strong positional promotion to establish macroscopic layouts. As generation progresses toward the terminal stage ( $t \rightarrow 0$ ),  $\kappa_{rs}(t)$  smoothly decays to 0, explicitly releasing the model from excessive spatial compression when micro-textures are crystallizing.

**Dynamic Ramp Function.** We use the scheduled coefficient  $\kappa_{rs}(t)$  to continuously modulate the transition boundaries that partition the frequency domain (as shown in Fig. 4):

$$\alpha_t = \alpha \cdot \kappa_{rs}(t), \quad \beta_t = \beta \cdot \kappa_{rs}(t), \quad (9)$$

where  $\alpha$  and  $\beta$  are base threshold hyperparameters. These dynamic boundaries dictate the effective range of positional extrapolation at any instantaneous timestep  $t$ .

To determine how each RoPE dimension is processed, recall from Sec. 2.2 that the  $i$ -th feature dimension relies on a base frequency  $\theta_i$ . The corresponding spatial wavelength of this rotation is mathematically defined as  $\lambda_i = 2\pi/\theta_i$ . Let  $s = L_{\text{target}}/L_{\text{train}}$  denote the target resolution promotion factor. We define the normalized spatial frequency ratio for the  $i$ -th dimension as:

$$r(i) = \frac{L_{\text{target}}}{\lambda_i} = \frac{L_{\text{target}}\theta_i}{2\pi}. \quad (10)$$

This ratio  $r(i)$  elegantly quantifies how many full wavelengths fit within the target sequence length. SHARP then rescales each RoPE

frequency through a time-dependent interpolation:

$$h_t(\theta_i) = (1 - \gamma(r(i), t)) \frac{\theta_i}{s} + \gamma(r(i), t) \theta_i, \quad (11)$$

where  $\gamma(r, t)$  is the dynamic ramp function defined by the boundaries  $\alpha_t$  and  $\beta_t$ :

$$\gamma(r, t) = \begin{cases} 0, & r < \alpha_t, \\ \frac{r - \alpha_t}{\beta_t - \alpha_t}, & \alpha_t \leq r \leq \beta_t, \\ 1, & r > \beta_t. \end{cases} \quad (12)$$

This mathematical design directly operationalizes the generative physics derived in Sec. 3.2. Equation 11 demonstrates that low-frequency modes ( $r < \alpha_t$ ) receive full linear extrapolation ( $\theta_i/s$ ), high-frequency modes ( $r > \beta_t$ ) are left perfectly uncompressed ( $\theta_i$ ), and intermediate modes undergo smooth interpolation.

Crucially, because  $\alpha_t$  and  $\beta_t$  decay with  $\kappa_{rs}(t)$ , the behavior is *highly dynamic*. In the early high-noise stage, the transition band is exceptionally wide, enforcing uniform spatial promotion across most frequencies to guarantee layout coherence. As denoising approaches the terminal detail-refinement phase, the boundaries  $\alpha_t$  and  $\beta_t$  shrink toward zero. This aggressively shifts the vast majority of frequency modes into the uncompressed regime ( $\gamma = 1$ ), thereby meticulously preserving the crisp high-frequency content essential for photorealistic RS imagery.

Algorithm 1 summarizes the complete SHARP inference procedure.

**Multi-scale Compatibility.** SHARP achieves independence from absolute pixel dimensions by operating on normalized timesteps  $t \in [0, 1]$  and dimensionless ratios  $r(i)$ . When scaling to a target resolution, all hyperparameters ( $\alpha_s, \alpha, \beta$ ) remain fixed; only the resolution promotion factor  $s = L_{\text{target}}/L_{\text{train}}$  is updated. For anisotropic targets,  $s$  is computed independently per spatial axis, with the same

*A satellite image of a rural market town with dense shop blocks, a bus station, surrounding crop fields, narrow feeder roads, and mixed residential and commercial parcels.*



*Satellite imagery showing a modern downtown beside a wide river, several bridges linking both banks, office towers casting long shadows, riverside parks, and dense commercial blocks arranged along the waterfront.*



**Figure 6: Multi-scale generation from a single prompt.** Each row demonstrates SHARP’s generation across six diverse resolutions and aspect ratios (from native  $1024 \times 1024$  up to  $3756 \times 2560$ ). SHARP maintains strict global semantic consistency across all scales while progressively crystallizing finer RS details at higher resolutions. This visual coherence validates that our dynamic RoPE scheduling successfully decouples macro-layout formation from high-frequency detail recovery.

**Table 1: Native-resolution comparison at  $1024 \times 1024$ . Higher is better.**

| Model          | CLIP↑        | Aes↑        | HPSv2↑       |
|----------------|--------------|-------------|--------------|
| FLUX (vanilla) | 25.53        | 5.18        | 0.236        |
| RS-FLUX (ours) | <b>28.62</b> | <b>5.97</b> | <b>0.283</b> |

ramp partitioning the corresponding RoPE dimensions. This "one-set-fits-all" configuration enables robust inference across diverse resolutions without per-resolution tuning, as validated in Sec. 4.

## 4 Experiments

### 4.1 Experimental Setup

**4.1.1 Training Data Construction.** We construct a domain-specialized corpus from the GeoChat dataset [13], which provides RS images with multi-turn dialogues. Since the question-answer format is poorly suited to caption-style supervision, we use Qwen-VL [2] to convert each sample into a compact prompt-image pair (Fig. 5b), yielding 102,952 pairs for RS-FLUX fine-tuning. The corpus spans multiple resolutions (Fig. 5a), promoting robustness to varying spatial extents.

**4.1.2 Evaluation Protocol.** We construct a benchmark of 100 GPT-5.4 generated prompts [25] covering diverse RS scenes. We evaluate generation quality using three reference-free metrics: CLIP Score [10] (semantic consistency), Aesthetic Score [23] (perceptual quality), and HPSv2 [32] (human preference).

**4.1.3 Multi-Scale Promotion Setting.** To evaluate generation beyond the  $1024 \times 1024$  training resolution, we test on three isotropic

targets ( $1536^2$ ,  $2048^2$ ,  $2560^2$ ) and three anisotropic targets ( $2048 \times 1536$ ,  $2560 \times 1920$ ,  $3072 \times 2048$ ). All methods generate images from the same 100 prompts under identical random seeds.

**4.1.4 Implementation Details.** RS-FLUX is fine-tuned from the  $1024^2$  FLUX checkpoint using AdamW ( $\beta_1=0.9$ ,  $\beta_2=0.999$ , weight decay 0.01) with a  $10^{-5}$  learning rate and cosine annealing. Training spans 10K steps with bf16 precision and an effective batch size of 64 ( $8 \times$  NVIDIA A6000 GPUs). For fair comparison, all inference methods share the same backbone, Euler sampler (28 steps, guidance scale 3.5), and seeds; only the RoPE adaptation strategy varies. SHARP hyperparameters ( $\alpha_s=3$ ,  $\alpha=1$ ,  $\beta=32$ ) follow the ablation results in Sec. 4.5.

### 4.2 RS-FLUX Validation at Native Resolution

Before evaluating resolution promotion strategies, we verify that domain-specialized fine-tuning improves generation quality at the native  $1024 \times 1024$  resolution. Table 1 compares vanilla FLUX with RS-FLUX on the same 100-prompt benchmark.

RS-FLUX improves over vanilla FLUX by +3.09 CLIP, +0.79 Aes, and +0.047 HPSv2. The largest gain appears on CLIP Score (+12.1% relative), indicating that full fine-tuning substantially narrows the semantic gap between generated outputs and RS-specific prompts—vanilla FLUX frequently misinterprets aerial terminology. The Aesthetic Score and HPSv2 improvements further confirm that RS-FLUX produces visually coherent aerial scenes with realistic color palettes and spatial layouts, establishing a stronger starting point for resolution promotion.

**Table 2: Multi-scale quantitative comparison under isotropic and anisotropic promotion. Higher is better for all metrics. The upper block evaluates isotropic promotion; the lower block evaluates anisotropic promotion. “Overall Avg.” averages all six resolutions.**

| Isotropic Promotion |              |             |              |              |             |              |              |             |              |                |             |              |
|---------------------|--------------|-------------|--------------|--------------|-------------|--------------|--------------|-------------|--------------|----------------|-------------|--------------|
| Method              | 1536 × 1536  |             |              | 2048 × 2048  |             |              | 2560 × 2560  |             |              | Isotropic Avg. |             |              |
|                     | CLIP↑        | Aes↑        | HPSv2↑       | CLIP↑        | Aes↑        | HPSv2↑       | CLIP↑        | Aes↑        | HPSv2↑       | CLIP↑          | Aes↑        | HPSv2↑       |
| RS-FLUX (baseline)  | 26.84        | 5.48        | 0.249        | 26.21        | 5.36        | 0.242        | 25.57        | 5.25        | 0.236        | 26.21          | 5.36        | 0.242        |
| PI                  | 27.18        | 5.56        | 0.255        | 26.73        | 5.46        | 0.248        | 26.09        | 5.33        | 0.241        | 26.67          | 5.45        | 0.248        |
| NTK-aware           | 27.44        | 5.63        | 0.261        | 27.02        | 5.54        | 0.255        | 26.46        | 5.42        | 0.248        | 26.97          | 5.53        | 0.255        |
| YaRN                | 27.71        | 5.78        | 0.269        | 27.31        | 5.69        | 0.263        | 26.78        | 5.57        | 0.257        | 27.27          | 5.68        | 0.263        |
| SHARP (ours)        | <b>28.03</b> | <b>5.89</b> | <b>0.276</b> | <b>27.72</b> | <b>5.82</b> | <b>0.271</b> | <b>27.28</b> | <b>5.71</b> | <b>0.266</b> | <b>27.68</b>   | <b>5.81</b> | <b>0.271</b> |

| Anisotropic Promotion |              |             |              |              |             |              |              |             |              |              |             |              |
|-----------------------|--------------|-------------|--------------|--------------|-------------|--------------|--------------|-------------|--------------|--------------|-------------|--------------|
| Method                | 2048 × 1536  |             |              | 2560 × 1920  |             |              | 3072 × 2048  |             |              | Overall Avg. |             |              |
|                       | CLIP↑        | Aes↑        | HPSv2↑       | CLIP↑        | Aes↑        | HPSv2↑       | CLIP↑        | Aes↑        | HPSv2↑       | CLIP↑        | Aes↑        | HPSv2↑       |
| RS-FLUX (baseline)    | 26.48        | 5.42        | 0.246        | 25.94        | 5.31        | 0.239        | 25.38        | 5.20        | 0.233        | 26.07        | 5.34        | 0.241        |
| PI                    | 26.96        | 5.50        | 0.252        | 26.42        | 5.40        | 0.246        | 25.88        | 5.28        | 0.239        | 26.54        | 5.42        | 0.247        |
| NTK-aware             | 27.25        | 5.58        | 0.259        | 26.74        | 5.48        | 0.252        | 26.19        | 5.36        | 0.245        | 26.85        | 5.50        | 0.253        |
| YaRN                  | 27.53        | 5.72        | 0.267        | 27.04        | 5.62        | 0.261        | 26.50        | 5.50        | 0.254        | 27.15        | 5.65        | 0.262        |
| SHARP (ours)          | <b>27.86</b> | <b>5.84</b> | <b>0.274</b> | <b>27.40</b> | <b>5.75</b> | <b>0.269</b> | <b>26.93</b> | <b>5.64</b> | <b>0.263</b> | <b>27.54</b> | <b>5.78</b> | <b>0.270</b> |

**Table 3: Ablation studies at 2048 × 2048. Higher is better. CL = CLIP Score, Aes = Aesthetic Score, HP = HPSv2. (a) Component contributions (FT = fine-tuning, SH = SHARP). (b) Scheduler form. (c) Scheduler coefficient  $\alpha_s$ . (d) Transition bounds ( $\alpha, \beta$ ).**

| (a) Component |    |              |             |             | (b) Scheduler form |              |             |             | (c) Coefficient $\alpha_s$ |              |             |             | (d) Bounds ( $\alpha, \beta$ ) |         |              |             |             |
|---------------|----|--------------|-------------|-------------|--------------------|--------------|-------------|-------------|----------------------------|--------------|-------------|-------------|--------------------------------|---------|--------------|-------------|-------------|
| FT            | SH | CL           | Aes         | HP          | Schedule           | CL           | Aes         | HP          | $\alpha_s$                 | CL           | Aes         | HP          | $\alpha$                       | $\beta$ | CL           | Aes         | HP          |
|               |    | 24.83        | 5.08        | .228        | Static             | 27.35        | 5.70        | .264        | 1                          | 27.48        | 5.74        | .266        | 1                              | 16      | 27.45        | 5.71        | .264        |
|               | ✓  | 26.15        | 5.48        | .253        | Linear             | 27.48        | 5.74        | .266        | 2                          | 27.61        | 5.78        | .269        | 1                              | 32      | <b>27.72</b> | <b>5.82</b> | <b>.271</b> |
| ✓             |    | 26.21        | 5.36        | .242        | Cosine             | 27.58        | 5.77        | .268        | 3                          | <b>27.72</b> | <b>5.82</b> | <b>.271</b> | 2                              | 32      | 27.61        | 5.77        | .268        |
| ✓             | ✓  | <b>27.72</b> | <b>5.82</b> | <b>.271</b> | Rational           | <b>27.72</b> | <b>5.82</b> | <b>.271</b> | 4                          | 27.55        | 5.75        | .267        | 2                              | 64      | 27.38        | 5.66        | .261        |

### 4.3 Quantitative Comparison

Table 2 compares SHARP with four baselines across six resolutions. The results show a consistent ranking: RS-FLUX (no RoPE adaptation) < PI < NTK-aware < YaRN < SHARP. Although PI, NTK-aware, and YaRN progressively improve the frequency treatment, they all remain static during denoising. By introducing dynamic scheduling, SHARP consistently outperforms the strongest static baseline.

**Overall performance.** SHARP achieves 27.54 CLIP, 5.78 Aes, and 0.270 HPSv2 on average, surpassing the strongest baseline YaRN by +0.39 / +0.13 / +0.008 and RS-FLUX by +1.47 / +0.44 / +0.029.

**Isotropic promotion.** SHARP improves over YaRN from 27.27 / 5.68 / 0.263 to 27.68 / 5.81 / 0.271. The margin *widens* with the promotion factor: +0.32 / +0.11 / +0.007 at 1536<sup>2</sup> versus +0.50 / +0.14 / +0.009 at 2560<sup>2</sup>. This widening corroborates the spectrum-aware design: as the promotion factor grows, static methods suffer increasingly severe high-frequency suppression, whereas SHARP’s dynamic scheduling compensates accordingly. From 1536<sup>2</sup> to 2560<sup>2</sup>, SHARP’s metrics degrade by 0.75 / 0.18 / 0.010 versus 0.93 / 0.21 /

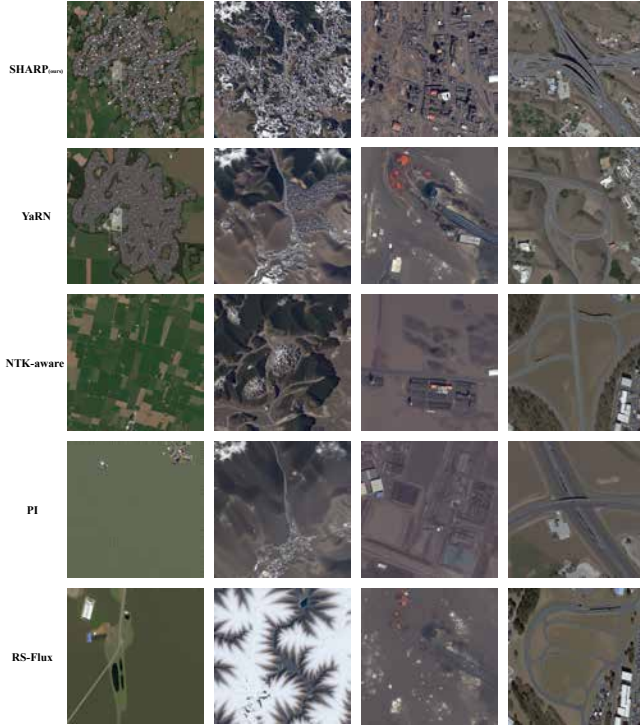
0.012 for YaRN, indicating slower quality decay under aggressive promotion.

**Anisotropic promotion.** SHARP leads at all anisotropic settings, with the largest margins at the most challenging 3072 × 2048 (26.93 / 5.64 / 0.263 vs. 26.50 / 5.50 / 0.254). Despite differing promotion factors along the two spatial axes, SHARP handles this gracefully through axis-independent RoPE rescaling via the dimensionless frequency ratio  $r(i)$ , consistent with its resolution-agnostic formulation.

### 4.4 Qualitative Comparison

Fig. 7 presents qualitative results at 2048 × 2048. Static baselines suffer from severe structural collapse, frequently degrading into hallucinated or overly smoothed homogeneous regions (e.g., the dense residential and industrial scenes). While YaRN roughly preserves macroscopic scene layouts, it heavily blurs detail-critical regions. In contrast, SHARP successfully retains both global coherence and high-frequency details. It synthesizes crisp building footprints, complex terrain textures, and multi-level highway topologies, robustly overcoming the textural degradation that plagues existing methods.





**Figure 7: Qualitative comparison at 2048×2048 resolution. Please zoom in for better detail visualization. Unlike static baselines that suffer from catastrophic over-smoothing of fine structures, SHARP consistently preserves crisp high-frequency features (e.g., dense buildings and road topologies) while maintaining superior global structural coherence.**

Fig. 6 demonstrates multi-scale consistency. Using the same prompt, SHARP generates RS images across six resolutions from  $1024 \times 1024$  to  $3756 \times 2560$ . Coherent spatial composition is maintained at all scales while finer details—vehicle contours, building boundaries, and road textures—progressively emerge as resolution increases. Spatial relationships remain stable even as the canvas expands and the aspect ratio changes, confirming that the Rational Decay Scheduler preserves global layout while the Dynamic Ramp Function progressively releases high-frequency modes. More zoomed-in detail comparisons and additional qualitative examples are provided in the supplementary material.

#### 4.5 Ablation Studies

We conduct ablation experiments at  $2048 \times 2048$  to validate each design choice. All ablations use the same 100-prompt benchmark. Table 3(a) confirms that fine-tuning and SHARP are complementary: each independently improves over vanilla FLUX, and their combination yields the best result (+2.89 CLIP). SHARP alone on vanilla FLUX (26.15 CLIP) already rivals RS-FLUX without SHARP (26.21 CLIP), demonstrating that dynamic positional adaptation provides considerable gains even without domain-specialized training. This complementarity arises because fine-tuning enriches the

**Table 4: Inference cost (seconds per image). Peak GPU memory is identical across all methods.**

| Method    | 1536 <sup>2</sup> | 2048 <sup>2</sup> | 2560 <sup>2</sup> |
|-----------|-------------------|-------------------|-------------------|
| RS-FLUX   | 66.7              | 130.8             | 235.4             |
| PI        | 67.1              | 131.2             | 237.1             |
| NTK-aware | 68.8              | 133.1             | 238.9             |
| YaRN      | 68.4              | 132.3             | 238.2             |
| SHARP     | 69.1              | 134.1             | 243.7             |

semantic prior while SHARP addresses the orthogonal positional OOD problem.

Panel (b) compares four scheduling strategies, all satisfying  $\kappa(1)=1$  and  $\kappa(0)=0$  with identical bounds  $(\alpha, \beta)=(1, 32)$ . Even the simplest dynamic schedule (linear) outperforms the static baseline by +0.13 CLIP, confirming that time-varying scheduling is universally beneficial. The rational form achieves the best results by maintaining near-unity promotion through the mid-stage before decaying rapidly near  $t=0$ , best matching the frequency-progressive recovery timeline.

Panel (c) reveals a unimodal optimum at  $\alpha_s=3$ . At  $\alpha_s=1$  (equivalent to linear decay), promotion decreases too gradually, leaving residual compression in the late stage; at  $\alpha_s=4$ , promotion is released too early, weakening mid-stage layout formation. The  $\alpha_s=1$  result (27.48 / 5.74 / .266) matches the linear entry in panel (b), providing independent cross-validation.

Panel (d) confirms  $(\alpha, \beta)=(1, 32)$  as optimal, coinciding with YaRN’s default ramp parameters [19]. This indicates that the frequency-domain partition is robust across modalities and that SHARP’s gains stem from the orthogonal temporal scheduling dimension. When  $\beta$  is too small ( $\beta=16$ ), most modes bypass promotion; when too large ( $\beta=64$ ), nearly all modes are uniformly interpolated, negating the frequency-selective advantage.

#### 4.6 Computational Efficiency

SHARP modifies only the RoPE frequency computation at each step and introduces no additional parameters. Table 4 reports inference time on a single A6000 GPU. SHARP adds less than 1.5% overhead relative to vanilla RS-FLUX across all tested resolutions. The extra cost comes only from evaluating Eq. 8–12 per step, involving element-wise arithmetic over the RoPE dimensions—negligible compared with the self-attention and feed-forward computations dominating each transformer pass. Peak GPU memory is unchanged because SHARP introduces no additional parameters or intermediate activations.

### 5 Conclusion

We present SHARP, a training-free resolution promotion framework for large-scale RS text-to-image synthesis. By replacing static extrapolation with a rational decay scheduler,  $\kappa_{rs}(t)$ , SHARP preserves both global layouts and dense high-frequency details. SHARP consistently outperforms static baselines across diverse scales with negligible overhead (< 1.5%). Future work includes content-adaptive schedules and sparse attention for extreme resolutions.



## References

- [1] Moeness Amin and Kamal Sarabandi. 2009. Special issue on remote sensing of building interior. *IEEE Transactions on Geoscience and Remote Sensing* 47, 5 (2009), 1267–1268.
- [2] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jing Zhou. 2023. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and more. *arXiv preprint arXiv:2308.12966* (2023).
- [3] Black Forest Labs. 2024. FLUX.1: A 12 Billion Parameter Rectified Flow Transformer for Text-to-Image Generation. <https://github.com/black-forest-labs/flux>.
- [4] bloc97. 2023. NTK-Aware Scaled RoPE allows LLaMA models to have extended (8k+) context size without any fine-tuning and minimal perplexity degradation. <https://www.reddit.com/r/LocalLLaMA/comments/14lz7j5/>.
- [5] Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. 2023. Extending context window of large language models via positional interpolation. *arXiv preprint arXiv:2306.15595* (2023).
- [6] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*. Ieee, 248–255.
- [7] Ruoyi Du, Dongliang Chang, Timothy Hospedales, Yi-Zhe Song, and Zhanyu Ma. 2024. Demofusion: Democratising high-resolution image generation with no \$\$\$ In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 6159–6168.
- [8] Muhammed Goktepe, Amir hossein Shamseddin, Erençan Uysal, Javier Muñelo Monteagudo, Lukas Drees, Aysim Toker, Senthil Asseng, and Malte Von Bloh. 2025. EcoMapper: Generative Modeling for Climate-Aware Satellite Imagery. In *Forty-second International Conference on Machine Learning*.
- [9] Yingqing He, Shaoshu Yang, Haoxin Chen, Xiaodong Cun, Menghan Xia, Yong Zhang, Xintao Wang, Ran He, Qifeng Chen, and Ying Shan. 2023. Scalecrafter: Tuning-free higher-resolution visual generation with diffusion models. In *The Twelfth International Conference on Learning Representations*.
- [10] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. 2021. Clipscore: A reference-free evaluation metric for image captioning. In *Proceedings of the 2021 conference on empirical methods in natural language processing*. 7514–7528.
- [11] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising diffusion probabilistic models. *Advances in neural information processing systems* 33 (2020), 6840–6851.
- [12] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang Wang, Weizhu Chen, et al. 2022. Lora: Low-rank adaptation of large language models. *Iclr* 1, 2 (2022), 3.
- [13] Kartik Kuckreja, Muhammad Sohail Danish, Muzammal Naseer, Abhijit Das, Salman Khan, and Fahad Shahbaz Khan. 2024. Geochat: Grounded large vision-language model for remote sensing. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 27831–27840.
- [14] Junyong Li, Keyan Chen, Liqin Liu, Zhengxia Zou, and Zhenwei Shi. 2025. Dual-branch GAN for cloud image generation based on cloud and background decoupling. *Chinese Space Science and Technology* 45, 5 (2025), 49–59.
- [15] Ke Li, Gang Wan, Gong Cheng, Liqiu Meng, and Junwei Han. 2020. Object detection in optical remote sensing images: A survey and a new benchmark. *ISPRS journal of photogrammetry and remote sensing* 159 (2020), 296–307.
- [16] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. 2023. Flow Matching for Generative Modeling. *arXiv:2210.02747* [cs.LG] <https://arxiv.org/abs/2210.02747>
- [17] Ruizhe Ou, Yuan Hu, Fan Zhang, Jiaxin Chen, and Yu Liu. 2025. GeoPix: Multi-Modal Large Language Model for Pixel-level Image Understanding in Remote Sensing. *arXiv:2501.06828* [cs.CV] <https://arxiv.org/abs/2501.06828>
- [18] William Peebles and Saining Xie. 2023. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*. 4195–4205.
- [19] Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. 2023. Yarn: Efficient context window extension of large language models. *arXiv preprint arXiv:2309.00071* (2023).
- [20] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. 2023. Sdxl: Improving latent diffusion models for high-resolution image synthesis. *arXiv preprint arXiv:2307.01952* (2023).
- [21] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10684–10695.
- [22] Chitwan Saharia, Jonathan Ho, William Chan, Tim Salimans, David J. Fleet, and Mohammad Norouzi. 2021. Image Super-Resolution via Iterative Refinement. *arXiv:2104.07636* [eess.IV] <https://arxiv.org/abs/2104.07636>
- [23] Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, et al. 2022. Laion-5b: An open large-scale dataset for training next generation image-text models. *Advances in neural information processing systems* 35 (2022), 25278–25294.
- [24] Ahmad Sebaq and Mohamed ElHelw. 2024. Rsdiff: Remote sensing image generation from text using diffusion model. *Neural Computing and Applications* 36, 36 (2024), 23103–23111.
- [25] Aaditya Singh, Adam Fry, Adam Perelman, Adam Tart, Adi Ganesh, Ahmed El-Kishky, Aidan McLaughlin, Aiden Low, AJ Ostrow, Akhila Ananthram, et al. 2025. Openai gpt-5 system card. *arXiv preprint arXiv:2601.03267* (2025).
- [26] Pierre Soille and Martino Pesaresi. 2002. Advances in mathematical morphology applied to geoscience and remote sensing. *IEEE Transactions on Geoscience and Remote Sensing* 40, 9 (2002), 2042–2055.
- [27] Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. 2020. Score-based generative modeling through stochastic differential equations. *arXiv preprint arXiv:2011.13456* (2020).
- [28] Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. 2024. Reformer: Enhanced transformer with rotary position embedding. *Neurocomputing* 568 (2024), 127063.
- [29] Jialu Sui, Yiyang Ma, Wenhan Yang, Xiaokang Zhang, Man-On Pun, and Jiaying Liu. 2024. Diffusion Enhancement for Cloud Removal in Ultra-Resolution Remote Sensing Imagery. *arXiv:2401.15105* [eess.IV] <https://arxiv.org/abs/2401.15105>
- [30] Datao Tang, Xiangyong Cao, Xingsong Hou, Zhongyuan Jiang, Junmin Liu, and Deyu Meng. 2024. CRS-Diff: Controllable Remote Sensing Image Generation with Diffusion Model. *arXiv:2403.11614* [cs.CV] <https://arxiv.org/abs/2403.11614>
- [31] Datao Tang, Xiangyong Cao, Xuan Wu, Jialin Li, Jing Yao, Xueru Bai, Dongsheng Jiang, Yin Li, and Deyu Meng. 2025. AeroGen: Enhancing Remote Sensing Object Detection with Diffusion-Driven Data Generation. *arXiv:2411.15497* [cs.CV] <https://arxiv.org/abs/2411.15497>
- [32] Xiaoshi Wu, Yiming Hao, Keqiang Sun, Yixiong Chen, Feng Zhu, Rui Zhao, and Hongsheng Li. 2023. Human preference score v2: A solid benchmark for evaluating human preferences of text-to-image synthesis. *arXiv preprint arXiv:2306.09341* (2023).
- [33] Xin Wu, Danfeng Hong, and Jocelyn Chanussot. 2021. Convolutional neural networks for multimodal remote sensing data classification. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2021), 1–10.
- [34] Chuang Yang, Bingxuan Zhao, Qing Zhou, and Qi Wang. 2025. MMO-IG: Multi-class and Multiscale Object Image Generation for Remote Sensing. *IEEE Transactions on Geoscience and Remote Sensing* 63 (2025), 1–12. doi:10.1109/TGRS.2025.3550404
- [35] Zhiping Yu, Chenyang Liu, Liqin Liu, Zhenwei Shi, and Zhengxia Zou. 2024. MetaEarth: A Generative Foundation Model for Global-Scale Remote Sensing Image Generation. *arXiv:2405.13570* [cs.CV] <https://arxiv.org/abs/2405.13570>
- [36] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. 2023. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF international conference on computer vision*. 3836–3847.
- [37] Bingxuan Zhao, Chuang Yang, Qing Zhou, and Qi Wang. 2025. RLI-DM: Robust Layout-Based Iterative Diffusion Model for SAR-to-RGB Image Translation. *IEEE Transactions on Geoscience and Remote Sensing* 63 (2025), 1–9. doi:10.1109/TGRS.2025.3613938